
Multi-Actor Multi-Critic Deep Deterministic Reinforcement Learning with a Novel Q-Ensemble Method

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Abstract

Reinforcement learning has gathered much attention in recent years due to its rapid development and rich applications, especially on control systems and robotics. When tackling real-world applications with reinforcement learning method, the corresponded Markov decision process may have huge discrete or even continuous state/action space. Deep reinforcement learning has been studied for handling these issues through deep learning for years, and one promising branch is the actor-critic architecture. Many past studies leveraged multiple critics to enhance the accuracy of evaluation of a policy for addressing the overestimation and underestimation issues. However, few studies have considered the architecture with multiple actors together with multiple critics. This study proposes a novel multi-actor multi-critic (MAMC) deep deterministic reinforcement learning method. The proposed method has three main features, including selection of actors based on non-dominated sorting for exploration with respect to skill and creativity factors, evaluation for actors and critics using a quantile-based ensemble strategy, and exploiting actors with best skill factor. Theoretical analysis proves the learning stability and bounded estimation bias for the MAMC. The present study examines the performance on a well-known reinforcement learning benchmark MuJoCo. Experimental results show that the proposed framework outperforms state-of-the-art deep deterministic based reinforcement learning methods. Experimental analysis also indicates the proposed components are effective. Empirical analysis further investigates the validity of the proposed method, and shows its benefit on complicated problems.

1 Introduction

Reinforcement learning (RL) has been studied for decades that is proved powerful when dealing with problems and applications which is assumed or is able to be formulated as a Markov decision process [25]. Numerous applications have been successfully solved by RL methods such as playing board games [26], training large-language model [24], and controlling humanoid [29]. RL methods are of several types, including value-based approach, policy gradient approach, policy optimization approach, and actor-critic approach [28]. This study focus on actor-critic based RL methods due to its nice performance on continuous control problems.

A common issue in RL is the huge or infinite space of states/actions, making conventional tabular methods inapplicable, and a straight forward solution is to construct approximation function for space transformation. As the rapid growth in high performance computing and deep learning [3], leveraging deep learning for building mapping function in RL methods, forming deep reinforcement learning (DRL), becomes possible. One representative method of DRL is the deep Q-learning (DQN)

Table 1: A compilation of some recent proposed actor-critic architectures according to the number of actors and critics

Method		#Critics		
		Single	Double	Multiple
#Actors	Single	DQN[23], DDPG[19].	TD3[9], SAC[10], OAC[7].	REDQ[6], MD3[34], QWPVOP[17].
	Double	-	DARC[21].	MA-TD3[18].
	Multiple	-	-	SUNRISE[16], MA-DDPG[18].

[23], which adopted deep convolution neural network to estimate the state-action function (a.k.a. Q function).

Advanced issues in deep reinforcement learning have been studied and investigated in past years [11]. Essential issues covers learning stability [2, 19, 9], estimation accuracy for handling issues of overestimation [4, 20], underestimation [7, 35] or both [32, 1, 15], sampling efficiency [10, 36, 22?], ensemble learning [21, 6, 16, 14, 34] and so forth, and hybridization of components for addressing these issues is proved to gain effectiveness and learning efficiency [12]. It is worth noting that these issues are highly correlated so that ensemble learning could handle estimation accuracy, which may bring learning stability and sampling efficiency, and thus results in better performance and convergence.

This study proposes a novel method: multiple-actors-multiple-critics (MAMC) deep deterministic reinforcement learning to address the above issues. The main features of the MAMC are threefold: 1) The MAMC manipulates multiple actors and critics in a current manner without predetermined relations, 2) The MAMC evaluates actors and critics as per a quantile-based ensemble strategy, and 3) The MAMC selects actors for exploration in learning on the basis of non-dominated sorting with respect to skill and creativity factors. The emerging MAMC is capable of facilitating nice exploration among multiple actors in the meantime improving and smoothing the learning of critics, which is key to stabilize the guiding force to actors.

The main contributions are listed as follows:

- Devise a parametric quantile-based ensemble estimator considering multiple actors and multiple critics for the target values of critics learning
- Design an actor evaluation and selection approach based on skill and creativity factors for exploration and exploitation
- Theoretically prove the MAMC has stable learning and bounded estimation bias
- Empirically examine the quality and validity of the MAMC, and investigate the run-time behavior of MAMC by inspecting into the proposed components

The rests of this study are organized as follows. Section 2 reviews recent RL methods under actor-critic architectures, and Section 3 introduces preliminaries of this study. Sections 4 and 5 in turn gives details and theoretical analysis for the proposed method. Section 6 examines the effectiveness for the proposed method. Section 7 draws conclusions.

2 Related Work

The actor-critic architecture is proposed by Konda and Tsitsiklis [13]. Table 1 compiles six out nine categories of actor-critic architectures in terms of the number of actors and critics for some recent proposed actor-critic-based RL methods. To the best of our knowledge, it is merely no study for actor-critic architectures with fewer number of actors than of critics.

SASC. For single-actor single-critic (SASC) architecture, a representative study is the deep Q-network (DQN) [23], which introduced deep convolution neural network into RL and brought nice performance on Atari. Another famous SASC-based method is the deep deterministic policy gradient (DDPG) [19]. DDPG ameliorated the learning stability and efficiency of DQN by combining deep learning with policy gradient for solving control problems with continuous action space.

75 **SADC.** Beyond SASC, lots of methods are proposed with a single actor and double critics, noted 1
 76 as SADC, for solving the issues of overestimation and exploration. In [9], a twin delayed deep
 77 deterministic policy gradient (TD3) was proposed. TD3 improved DDPG by adopting two critic
 78 networks, where a minimum of the corresponded two target networks are served as the computa-
 79 tional basis of target value. TD3 also proposed the delayed update of actor, i.e., a lower update
 80 frequency than critics, for stabilizing the learning of the actor. Soft Actor-Critic (SAC) considered
 81 stochastic policy and introduced soft value function for training the two critics of soft Q-function
 82 [10]. Specifically, SAC trained a stochastic policy network to transform noise to an action for a given
 83 state as condition, and the training depends on a policy gradient for maximizing the randomness of
 84 the resulting actions, and the approximated Q values obtained from the minimum of the two critics
 85 as TD3. Different from TD3, SAC trained the two critics independently according to the target soft
 86 value function network, which is soft-updated by the soft value function network, while the soft
 87 value function network is trained by minimizing the different to the target value, which is calculated
 88 as the expectation of state-action value of the minimum of the two critics over action given by the
 89 actor. Optimistic Actor-Critic (OAC) further pointed out the issues of inefficient exploration owing
 90 to insufficient pessimistic in TD3 and SAC, and proposed an amelioration to guide the exploration
 91 according to the approximated lower and upper bounds of the state-action value function [7].

92 **SAMC.** From the observation of improvement from SASC to SADC, many methods considered 2
 93 increasing the number of critics for improving the estimation accuracy, forming the single-actor
 94 multi-critic architecture (SAMC). Randomized ensembled double Q-learning (REDQ) [6] estimated
 95 the state-action value using the same strategy of minimum the same as TD3, yet the two critics
 96 were randomly selected from a pool of critics. REDQ also introduced a high update-to-date (UTD)
 97 ratio of 20 to address the issue of sample efficiency. For addressing the estimation accuracy issue,
 98 quasi-median Q-learning (QMQ) used the quasi-median among multiple state-action values, each
 99 of which from a critic, to estimate the state-action value, and applied on TD3, forming the QMD3.
 100 The QMD3 trained actor with delay the same as TD3, but each update is guided by all critics rather
 101 than a single one for exploration improvement. Weakly pessimistic value estimation and optimistic
 102 policy optimization (WPVOP) [17] proposed weakly pessimistic value estimation and optimistic
 103 policy optimization; the former increased and smoothed the lower confidence bound, whilst the latter
 104 encourages and increases the state-action value, as the maximum action of minimum state-action
 105 values, if the distribution of state-action values with different actions on a given state is centralized,
 106 i.e., the standard deviation less than some threshold.

107 **DADC.** From single actor to double actors, double actors and regularized critics (DARC) [21] adopted 3
 108 double actors as well as double critics (DADC) and proposed soft target value as a linear combination
 109 of the minimum and maximum state-action values of the two actions given by two target actors, each
 110 of which is a minimum over two target critics. DARC revised the loss function adopted in TD3 by
 111 introducing a weighted regularization term of cross-critic error, i.e., the difference between the two
 112 critics.

113 **DAMC.** A multi-actor mechanism (MAM) was proposed in [18]. The MAM was composed of two 4
 114 components, i.e., obsolescence technology (OT) and Q-value weighting technology (WT). For each
 115 actor, the OT adaptively adjust the probabilities of interacting with environment according to the
 116 accumulated reward in an episode. For each critic, the WT estimated the target of state-action value
 117 by a linear combination of state-action values achieved by the corresponding actor, and by the other
 118 actors. The MAM was integrated into TD3 (MA-TD3) with double actors and four critics, forming
 119 the double-actor multi-critic architecture (DAMC).

120 **MAMC.** For multi-actor multi-critic (MAMC) architecture, the MAM was also integrated into 5
 121 DDPG (MA-DDPG) with three actors and three critics, which became another MAMC-based method
 122 [18]. An early MAMC method is the Simple UNified framework for Reinforcement learning using
 123 enSembles (SUNRISE) [16]. SUNRISE manipulated multiple SAC agents, each contained a pair of
 124 soft Q-function and an actor. SUNRISE integrated weighted Bellman backup, which decreases the
 125 influence from high variance transitions, and upper confidence bound (UCB) exploration [5].

126 3 Preliminaries 6

127 Given a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, a state transition 7
 128 probability $\mathcal{P}_{s,s'}^a$, a reward function $\mathcal{R}_{s,a} = \mathbb{E}[R_{t+1}|S_t = s, A_t = a]$, and a discount factor γ ,

Table 2: Notation system 1

Symbol	Meaning
N_A, N_C, N_B	Number of actors, critics, and mini-batch size
π_ϕ	Actor network with parameter ϕ
$A, \tilde{A}$	Actors and selected actors
C, C'	Critics and target critics
$Q_\theta (Q_{\theta'})$	Critic (target) network with parameter $\theta (\theta')$
$\mathcal{R}, \mathcal{B}$	Replay buffer and mini-batch
M	Sample multiple reuse
(s, a, r, s')	Transition from state s to next state s' by action a with reward r
γ	Discount factor
$\vec{J}_s(A), \vec{J}_c(A)$	Skill and creativity factors of actors A
$\prec$	Crowded-comparison operator
$\mathcal{N}(\mu, \sigma)$	Gaussian distribution with mean μ and variance σ^2
τ	Soft update ratio

reinforcement learning aims at learning policy π to achieve optimal return from rewards. A famous method is the Q-learning [33], which learns a state-action value function for estimating the reward function $\mathcal{R}_{s,a}$

$$\begin{aligned} Q^\pi(s, a) &= \mathbb{E}[R_{t+1} | S_t = s, A_t = a] \\ &= \mathbb{E}[r_{t+1} + \gamma Q^\pi(S_{t+1} = s', A_{t+1} = \pi(s_{t+1})) | S_t = s, A_t = a]. \end{aligned} \quad (1)$$

The estimation forms a Bellman equation, which can be solved by temporal difference (TD) [27, 30] methods. TD methods approximate the expected return by gradually lowering down the TD error, i.e., the difference of returns between the state-action value $Q(s, a)$ and the TD-target $r_{t+1} + \gamma V(s_{t+1})$, where $V(s_{t+1})$ is the state-value function satisfying $V(s_{t+1}) = Q(s_{t+1}, \pi(s_{t+1}))$.

Establishing approximation function to form a mapping from state space to action space $\pi_\phi : \mathcal{S} \rightarrow \mathcal{A}$ and a mapping from state space and action space to a real-value $Q_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ by deep neural network forms deep reinforcement learning. According to [9], the update of critic then can be made by minimizing the critic loss function:

$$J_Q(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{B}}[(Q_\theta(s, a) - r + \gamma V_\phi(s'; \theta'))^2], \quad (2)$$

subject to

$$V_\phi(s'; \theta') = Q_{\theta'}(s', \pi_\phi(s')) + \epsilon, \quad (3)$$

where θ' is the parameters of critic target with soft update, satisfying $\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$, and ϵ is the policy noise similar to the technique adopted in SARSA learning [28]. The soft update is for stabilizing the learning of critic network using a fixed target. Then, the update of actor is to minimize the actor loss function:

$$J_\pi(\phi; \theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{B}}[-Q_\theta(s, \phi(s))]. \quad (4)$$

4 MAMC 12

This study proposes a multi-actor-multi-critic architecture-based RL method: the Multi-Actor Multi-Critic deep deterministic reinforcement learning (MAMC). There are three main features in the proposed MAMC, including the adoption of multiple actors and critics without predefined interaction, the quantile-based ensemble estimation, and the selection of actors as per proposed skill and creativity factors for exploration and exploitation. Table 2 provides the notation system used in this study.

4.1 The Overall Procedure 14

Algorithm 1 gives the main procedure of the proposed MAMC. At initialization, the MAMC generates a set of N_A actor networks A and a set of N_C critic networks C with random parameters, and set the parameters of each target network according to the parameters of its corresponded critic network. The replay buffer $\mathcal{R}$ is also initialized by random actions of a predefined size. During each iteration, there are three main stages: critics learning stage, actors learning state, and exploration stage.

Algorithm 1 Main procedure of MAMC

```

1: Initialize a set of  $N_A$  actor networks  $A$  with random parameters  $\{\phi_i\}_{1 \leq i \leq N_A}$ 
2: Initialize a set of  $N_C$  critic networks  $C$  with random parameters  $\{\theta_j\}_{1 \leq j \leq N_C}$ 
3: Initialize a set of  $N_C$  target networks  $C'$  with critics  $\theta'_j \leftarrow \theta_j$  for  $1 \leq j \leq N_C$ 
4: Initialize replay buffer  $\mathcal{R}$ 
5: while Not Terminated do
6:    $\triangleright$  Critics Learning
7:    $\{\mathcal{B}_j\}_{1 \leq j \leq N_C} \sim \mathcal{R}$   $\triangleright$  Sample a mini-batch from replay buffer  $R$  for each critic
8:   for  $m \leftarrow 1$  to  $M$  do  $\triangleright$  Sample multiple reuse
9:     Update  $\theta_j$  on  $\mathcal{B}_j$  according to Eqs. (6) and (7) for  $1 \leq j \leq N_C$ 
10:    Update  $\theta'_j$  by soft update for  $1 \leq j \leq N_C$ 
11:   end for
12:    $\triangleright$  Actors Learning
13:    $\{\mathcal{B}_i\}_{1 \leq i \leq N_A} \sim \mathcal{R}$   $\triangleright$  Sample a mini-batch from replay buffer  $R$  for each actor
14:   for  $m \leftarrow 1$  to  $M$  do  $\triangleright$  Sample multiple reuse
15:      $j \leftarrow m \bmod N_C$   $\triangleright$  Guided by each critic in turn
16:     Update  $\phi_i$  by  $\theta_j$  on  $\mathcal{B}_i$  according to Eq. (4) for all  $1 \leq i \leq N_A$ 
17:   end for
18:    $\triangleright$  Exploration
19:    $\tilde{A} \leftarrow \text{Selection}(\vec{J}_s(A; C), \vec{J}_c(A; C), \prec)$   $\triangleright$  Crowded-comparison operator
20:    $(r, s') \leftarrow \text{Env}(s, a = \pi_{\phi \sim \tilde{A}}(s) + \epsilon)$ ,  $\epsilon \sim \mathcal{N}(0, \sigma)$   $\triangleright$  Interact with environment
21:    $\mathcal{R} \leftarrow \mathcal{R} \cup (s, a, r, s')$ 
22:    $\pi^* \leftarrow \arg \min_{\phi} J_s(\phi; C)$ 
23: end while
24: return  $\pi^*$ 

```

4.2 Quantile-based Ensemble Estimation

In critics learning stage, N_C sets of mini-batch $\{\mathcal{B}_j\}_{1 \leq j \leq N_C}$ are sampled from the replay buffer $\mathcal{R}$, and each critic is trained on a specific mini-batch for M times for improving the stability.

Definition 1. For each transitions $(s, a, r, s') \in \mathcal{B}_j$, the TD-target for j th critic Q_{θ_j} is defined as the median action of the q th-quantile among the critic targets:

$$y(s, a) = r + \gamma \hat{V}_A(s'), \quad (5)$$

subject to

$$\begin{aligned} \hat{V}_A(s'; C') &= \text{Med}(\{\hat{V}_{\phi_i}(s'; C')\}_{1 \leq i \leq N_A}) \\ \hat{V}_{\phi_i}(s'; C') &= \text{Quantile}_q(\{Q_{\theta'_j}(s', \pi_{\phi_i}(s') + \epsilon)\}_{1 \leq j \leq N_C}). \end{aligned} \quad (6)$$

The critic loss function is therefore defined as

$$J_Q(\theta_j; C') = \mathbb{E}_{(s, a, r, s') \sim \mathcal{B}}[(Q_{\theta_j}(s, a) - r + \gamma V_A(s'; C'))^2]. \quad (7)$$

All the target critics are soft-updated with parameter τ after one out of M iterations of training, which is capable of sharing information to each target critic from all the other critic targets and bring to the next iteration.

For the learning of actors, the MAMC also sampled N_A sets of mini-batch $\{\mathcal{B}_i\}_{1 \leq i \leq N_A}$ from the replay buffer $\mathcal{R}$ as it does in critics learning stage. The training of each actor π_{ϕ_i} is in turn guided by each critic Q_{θ_j} with objective $J_{\pi}(\phi_i; \theta_j)$ (cf. Eq. (4)) on its mini-batch $\mathcal{B}_i$. The idea of updating M times within a mini-batch for each actor and critics is similar to sample multiple reuse (SMR) proposed in [22], which is able to stabilize the learning sequence.

4.3 Actor Evaluation, Exploration, and Exploitation

After training of actors and critics, the exploration stage is to select appropriate actors for interacting with the environment. The evaluation of an actor π_{ϕ_i} is based on two factors, i.e., skill and creativity, both are determined by the ensemble estimation of state value function.

176 **Definition 2.** Ensemble estimation of state value function is defined as the q th-quantile of state-action 1
 177 value function over critics C :

$$\hat{V}_{\phi_i}(s^{(k)}; C) = \text{Quantile}_q(\{Q_{\theta_j}(s^{(k)}, \pi_{\phi_i}(s^{(k)}))\}_{1 \leq j \leq N_C}), \quad 2 \quad (8)$$

178 where $s^{(k)}$ is the k th transition in a mini-batch. The consideration of skill factor guarantees the quality 3
 179 of interaction, whilst the consideration of creativity factor preserves the diversity of interaction.

180 The skill factor evaluates the optimality of an actor through the scoring ability on the ensemble 4
 181 estimation

$$J_s(\phi_i; C) = N_B^{-1} \sum_{k=1}^{N_B} \hat{V}_{\phi_i}(s^{(k)}), \quad 5 \quad (9)$$

182 while the creativity factor examines the diversity of an actor on the critics through the closeness of 6
 183 each critic to the ensemble estimation with respect to mean absolute error

$$J_c(\phi_i; C) = N_B^{-1} N_C^{-1} \sum_{k=1}^{N_B} \sum_{j=1}^{N_C} |Q_{\theta_j}(s^{(k)}, \pi_{\phi_i}(s^{(k)})) - \hat{V}_{\phi_i}(s^{(k)})|. \quad 7 \quad (10)$$

184 Both factors are expectation over a mini-batch. Note that the two factors depends on all the critics as 8
 185 rather than critic targets since the actors are guided by critics. The selection of actors on the two factor
 186 hinges upon the crowd-comparison operator [8] by considering the two factors as two objective
 187 values. The top- $\sqrt{N_A}$ actors $\tilde{A}$ are selected, which serves as the candidate actors for interaction
 188 with the environment. Specifically, an actor is randomly picked from the candidate actors $\tilde{A}$ for
 189 determining a single step of interaction with the environment. The MAMC also records an optimal
 190 policy with highest skill factor for exploitation at each iteration; that is, the MAMC only returns a
 191 single actor for inference due to the efficiency in terms of time and space complexity.

192 5 Theoretical Analysis 9

193 This section gives some nice properties for the MAMC. First, the target values obtained by multiple 10
 194 actors are more stable in terms of variance than using a single actor.

195 **Theorem 1.** *The variance of target values obtained by multiple actors are less than that using a 11
 196 single actor:*

$$\mathbb{V}[\hat{V}_A(s'; C')] \leq \mathbb{V}[\hat{V}_\phi(s'; C')] \quad 12 \quad (11)$$

197 Similarly, the target values obtained by multiple critics are more stable than using a single critic. 13

198 **Theorem 2.** *The variance of target values obtained by multiple critics are less than using a single 14
 199 critic.*

$$\mathbb{V}[\hat{V}_\phi(s'; C')] \leq \mathbb{V}[\hat{V}_\phi(s'; \theta')] \quad 15 \quad (12)$$

200 Thus, the learning stability of the MAMC, with lowest variance, is greater than SAMC and SASC. 16

201 Further, this study investigate the property of estimation error, which is a good metric for indicating 17
 202 the estimation accuracy [21].

203 **Definition 3.** The estimation error of MAMC is defined as the difference between expectation of 18
 204 estimate values and the expectation of optimal policy π .

$$\mathcal{E}_{A,C} = \mathbb{E}[\hat{V}_A(s'; C)] - \mathbb{E}[\hat{V}_\phi(s'; C)] \quad 19 \quad (13)$$

205 Then the MAMC holds the following properties. 20

206 **Theorem 3.** *The estimation error of MAMC is between the estimation error of multiple actors with 21
 207 minimum and maximum critics.*

$$\mathcal{E}_{A, Q_{\theta_{\min}}} \leq \mathcal{E}_{A,C} \leq \mathcal{E}_{A, Q_{\theta_{\max}}} \quad 22 \quad (14)$$

208 **Theorem 4.** *The estimation error of MAMC is between the estimation error of multiple critics with 23
 209 minimum and maximum actors.*

$$\mathcal{E}_{\pi_{\phi_{\min}}, C} \leq \mathcal{E}_{A,C} \leq \mathcal{E}_{\pi_{\phi_{\max}}, C} \quad 24 \quad (15)$$

210 Hence, the estimation error of MAMC is in between the maximum and minimum of SAMC and 25
 211 MAMC. The proofs of above theorems will be given in supplementary material Section B due to space
 212 limitation.

Table 3: Wilcoxon signed rank test for TD3 and DARC compared with the MAMC at early (100k), middle (200k), and late stage (300k). The win/tie/lose denotes the number of environments that the MAMC is significantly superior, equal, and inferior to the corresponding test method.

Stage (win/tie/lose)	TD3-SMR	DARC-SMR	SAC-SMR	REDQ-SMR
100k	3/2/0	2/2/1	3/2/0	0/4/1
200k	2/3/0	1/4/0	2/2/1	0/4/1
300k	2/3/0	1/3/1	1/4/0	0/3/2

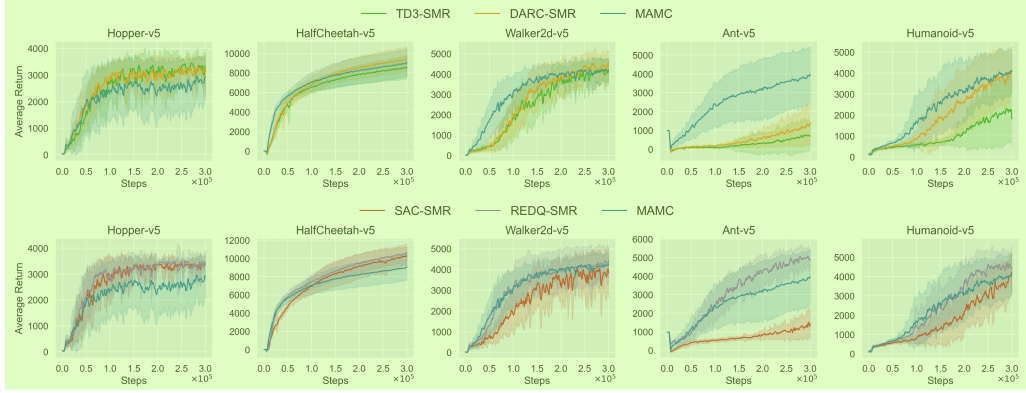


Figure 1: Average return against environment steps for TD3-based and SAC-based methods by comparison with the MAMC on the five environments

6 Experimental Results

This section examines the performance of the proposed MAMC method in terms of effectiveness and efficiency through experiments. Further analysis is made for showing the effectiveness of the proposed components, the sensitivity of introduced hyperparameters, and the validity of the MAMC.

6.1 Experimental Settings

The experiments are conducted on a set of five test environments chose from the well-known MuJuCo benchmark [31], including Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, and Humanoid-v5. These environments are all have continuous state and action spaces with different dimensions. Regarding the dimensionality, the difficulty of each environment can be regarded as either simple (Hopper-v5), medium (HalfCheetah-v5, Walker2d-v5), or hard (Ant-v5, Humanoid-v5). The properties of these environment can be found in the supplementary material Section A.3.

This study selects four state-of-the-art RL methods for performance comparison, including two with deterministic policy: TD3[9] and DARC[21], and two with stochastic policy: SAC[10], REDQ[6]. Both TD3 and SAC used a single actor with two critics. REDQ also adopts a single actor but with ten critics, while DARC exploits two actors and two critics. The proposed MAMC utilizes ten actors and ten critics. An analysis on the number of actors and critics can be found in the supplementary material Section C.3. In addition, as the MAMC considers sample multiple reuse (SMR) [22], all the four test methods are implemented as SMR versions, which are reported with better performance than the original versions, for a fair comparison.

The hyperparameter settings for the four baseline methods follow their original suggestions. The termination criterion is set to 300k environmental steps. All experiments conducted 10 trials, and each trial is an average over twenty seeds for return if not stated. All figures are uniformly smoothed. For significance analysis, this study adopts the Wilcoxon ranksum test with .05 significant level. The error bars are within the range $[\mu - \sigma, \mu + \sigma]$, which are generated by standard deviations with the assumption of normally distributed errors. For more details about the experimental settings, please refer to the supplementary material Section A.

Table 4: Average and standard deviation of return for the MAMC with single-objective and multi-objective actor selection strategies on Ant-v5 over eight trials at early (100k), middle (200k), and late stage (300k). The bold symbol implies the highest value.

Stage	100k		200k		300k	
Ant-v5	SO	MO	SO	MO	SO	MO
Mean	1980	2701	2805	3611	3395	4276
Std.	800	1235	1014	1631	1278	1260

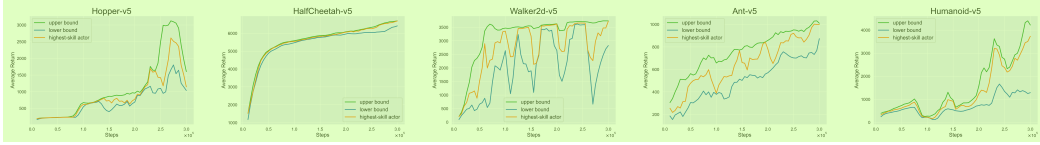


Figure 2: Average return for the best, worst, and skilled (selected) actors in the MAMC in a specific trial on the five test environments

6.2 Effectiveness

Table 3 compares the Wilcoxon signed rank test for TD3 and DARC compared with the MAMC at early (100k), middle (200k), and late stage (300k). The details are provided in supplementary material Section C.1. At early stage, the MAMC achieves better quality than the two deterministic methods TD3-SMR and DARC-SMR. Comparing to the two SAC-based methods, the MAMC outperforms SAC-SMR but performs slightly worse than REDQ-SMR on the Hopper-v5 environment. At middle stage, the MAMC still betters TD3-SMR and DARC-SMR, yet the improvement becomes smaller than that at early stage. As for the two SAC-based methods, the trend on REDQ-SMR keeps, while the improvement on SAC-SMR also decreases. At late stage, the lead to TD3-SMR, DARC-SMR, and SAC-SMR further shrinks that the MAMC is slightly superior to TD3-SMR and SAC-SMR, but is comparable to DARC-SMR. The REDQ-SMR further surpasses the MAMC on Humanoid-v5 environment. These results reflect the merits of MAMC at early and middle stage, and the demerit at late stage.

6.3 Efficiency

Figure 1 draws the average return against environment steps for TD3-based and SAC-based methods by comparison with the MAMC on the five environments. Compared with TD3-based methods, the MAMC gains faster convergence on the three more complicated environments, i.e., Walker2d-v5, Ant-v5, and Humanoid-v5. Similarly, the MAMC converges faster than SAC-SMR on these three environments, yet the REDQ-SMR converges nicer than the MAMC on all except Walker2d-v5. These results validate the efficiency of the MAMC against the two deterministic method TD3-SMR and DARC-SMR, and the simpler stochastic method SAC-SMR.

6.4 Components Analysis

Table 4 lists the average and standard deviation of return for the MAMC with single-objective (MAMC-SO) and multi-objective actor selection strategies on Ant-v5 over eight trials. The MAMC-SO averages the skill and creativity factors and selects the top actors by sorting for exploration. The exploitation selection mechanism for MAMC-SO and MAMC is the same. From the table, the MAMC performs better than MAMC-SO at all the three stages, which verifies the effectiveness of the proposed multi-objective actor selection mechanism.

Figure 2 plots the average return for the best (upper bound), worst (lower bound), and skilled (selected) actors in the MAMC in a specific trial on the five test environments. On HalfCheetah-v5 and Humanoid-v5, the MAMC is capable of selecting good actor approaching the upper bound, to wit, the best actor. For Hopper-v5, Walker2d-v5, and Ant-v5, the MAMC tracks the moving upper bound, and in most of the time the selected actor having quality beyond the average of upper and

Table 5: Average and standard deviation of return for the MAMC with different quantile parameter q

q	= 0.1	= 0.2	= 0.3	= 0.4	= 0.5
HalfCheetah-v5	9119±1077	8153±1115	9117±1070	9191±1043	9466±1256
Walker2d-v5	3188±1516	4324±1038	4083±927	3385±1039	1406±561

Figure 3: Estimated and ground-truth average discounted return against environment steps for the MAMC with different quantile parameters q on HalfCheetah-v5 and Walker2d-v5

lower bounds. These results validate the effectiveness of the proposed skill factor for actor selection for exploitation.

6.5 Sensitivity Analysis

Figure 3 plots the estimated and ground-truth average discounted return against environment steps for MAMC with different quantile parameters q on HalfCheetah-v5 and Walker2d-v5. It is obvious that the estimated value increases as q increases; the best q in terms of the smallest distance to ground-truth value is 0.4 for HalfCheetah-v5 and 0.3 for Walker2d-v5. However, the values are inconsistent to the best q in terms of the average return, which is 0.5 for HalfCheetah-v5 and 0.2 for Walker2d-v5 (cf. Table 5). That is, the setting of quantile parameters q should also consider the environmental preferences of optimism and pessimism. In general, a range between 0.2 and 0.3 is a good setting for environments which favor pessimism, and a value between 0.3 and 0.4 is nice for optimism cases; thus, a robust q value may near 0.3, but the best one for a specific environment still needs to be investigated.

6.6 Validity Analysis

The proposed MAMC is based on deterministic policy, and the results have shown that the MAMC can ameliorate the performance of TD3-SMR and REDQ-SMR. The MAMC is also beneficial in comparison to SAC-SMR, a simple but powerful method with stochastic policy. Past studies have discovered the potential of stochastic policy over deterministic policy, and this may be the weakness of the MAMC, which is considered as the main reason to be surpassed by REDQ-SMR.

7 Conclusions

This study proposes a multi-actor multi-critic deep deterministic reinforcement learning method. The MAMC includes a selection of actors for exploration using skill and creativity factors, an ensemble target value based on a predefined quantile parameter, and a selection of best actor regarding skill factor for exploitation. Theoretical analysis proves the MAMC having bounded estimation error, and learning stability over SAMC and MASC. From experimental results, the MAMC excels TD3-SMR, DARC-SMR, and SAC-SMR with better quality and faster convergence on the selected environments in MuJoCo. The validity analysis shows a weakness of deterministic based method and is also a possible future extension. Another promising orientation for future research is to adapt the quantile parameter to address the issue of estimation accuracy by balancing optimism and pessimism.

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